

# AI-POWERED INSURANCE CLAIMS INTELLIGENCE ASSISTANT

## Overview

Insurance teams process large volumes of claims containing customer information, policy details, claim descriptions, incident information, amounts, supporting documents, and claim-status information. Reviewing these records manually can be time-consuming, particularly when teams need to identify potentially high-risk claims, understand claim history, and prioritize cases for further investigation.

Users may ask questions such as:

- “Which claims require further investigation?”
- “Identify claims with unusual claim amounts or patterns.”
- “Which claims have characteristics similar to previously investigated claims?”
- “What information supports the risk assessment for this claim?”
- “Summarize the key details of this claim for an insurance reviewer.”
- “Which claims should be prioritized for manual review?”

Build an **AI-Powered Insurance Claims Intelligence Assistant** that combines claims-data analysis, similarity search, anomaly/risk detection, claim summarization, natural-language querying, and AI-generated insights to help insurance teams review and investigate claims more efficiently.

## Tasks

| Task                                                                        | Task Overview                                                                                                                  | Weightage   |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Task 1 – Claims Data Preparation &amp; Knowledge Base</b>                | Prepare insurance claims data and build the foundation for intelligent claims analysis.                                        | <b>20%</b>  |
| <b>Task 2 – Claims Analysis, Similarity Search &amp; Risk Detection</b>     | Analyze claims and identify unusual or potentially high-risk patterns.                                                         | <b>25%</b>  |
| <b>Task 3 – Multi-Agent Claims Intelligence &amp; Investigation Support</b> | Extend the claims analysis system into a specialized multi-agent workflow for deeper claim investigation and decision support. | <b>25%</b>  |
| <b>Task 4 – Evaluation, Claims Reporting &amp; User Application</b>         | Evaluate the claims-intelligence capabilities and provide a simple interface for insurance users.                              | <b>20%</b>  |
| <b>Task 5 – Architecture, Documentation &amp; Demonstration</b>             | Document the complete solution and demonstrate the working Insurance Claims Intelligence Assistant.                            | <b>10%</b>  |
| <b>Total</b>                                                                |                                                                                                                                | <b>100%</b> |

## Task Details

### Task 1 – Claims Data Preparation & Knowledge Base

- Prepare the insurance claims data for intelligent analysis. The solution should process relevant information such as:
  - Customer/policy information
  - Claim details
  - Incident information
  - Claim amount
  - Claim status
  - Claim history
  - Claim descriptions, where available
- The participant should create appropriate analytical features and prepare claim text for semantic retrieval where applicable.
- Claim identifiers and source information should be retained so that generated insights can be traced back to the underlying records.

### Task 2 – Claims Analysis, Similarity Search & Risk Detection

- Build the core claims-intelligence capability.
- The solution should allow an insurance reviewer to search for claims similar to a particular claim or natural-language description.
- The system should identify:
  - Unusual claim amounts
  - Repeated or unusual claim patterns
  - Relevant claim-history indicators
  - Similar historical claims
  - Potentially high-risk characteristics
  - Claims requiring further review
- A clearly documented risk-scoring or prioritization approach should be implemented.
- The system should provide supporting evidence for important risk indicators rather than presenting unexplained risk scores.

### Task 3 – Multi-Agent Claims Intelligence & Investigation Support

- Extend the claims-analysis system into a specialized multi-agent workflow.
- The implementation should include:
  - **Claims Retrieval Agent** – retrieves similar historical claims and relevant information.
  - **Claims Risk Analysis Agent** – analyzes claim characteristics and identifies risk indicators.
  - **Anomaly Detection Agent** – investigates unusual claim patterns.
  - **Claims Summarization Agent** – generates a concise summary of the claim and relevant history.

- **Investigation Support Agent** – consolidates the findings and identifies information that may require additional human review.
- The system should also demonstrate:
  - Agent handoff for complex claims.
  - **A2A communication** where applicable.
  - Evidence-backed claims analysis.
  - Traceability between AI-generated findings and the underlying claim records.
- The system should support **human review and decision-making** rather than presenting an AI-generated assessment as an automatic final insurance decision.

#### **Task 4 – Evaluation, Claims Reporting & User Application**

Evaluate the quality of the claims-intelligence system. The evaluation should consider:

- Similar-claim retrieval relevance
- Risk/anomaly detection performance
- Claim-summary accuracy
- Evidence grounding
- Natural-language query accuracy
- Consistency of generated insights

The front-end should allow an insurance reviewer to:

1. Search or select a claim.
2. View claim information.
3. Retrieve similar historical claims.
4. View risk indicators.
5. Review the claim summary.
6. View supporting evidence.
7. Identify claims requiring further manual investigation.

#### **Task 5 – Architecture, Documentation & Demonstration**

- Provide complete technical documentation and demonstrate the working solution.
- The architecture should clearly show:  
**Claims Data → Data Processing → Retrieval/Analytics → Risk & Anomaly Detection → Multi-Agent Intelligence → Claims Insights → Human Reviewer**
- The documentation should explain the major technical decisions, analytical approaches, agent workflow, evaluation methodology and design trade-offs.

#### **Deliverables**

##### **1. Architecture Diagram**

- Include a high-level view of your system in JPEG or PDF.
- Show how data is processed from the provided insurance claims dataset through to the user's query and final claims-intelligence response.

##### **2. Design**

- Articulate system design decisions taken and the trade-offs made.

### 3. Full Executable Code (Microservice)

- Focus on clarity and modularity.
- Provide a README explaining:
  - Project setup (how to install or run)
  - Sample usage (e.g., a claim query and the resulting analysis/insights)

### 4. Panel Presentation (10 minutes)

- Demo of your working solution outlining the design and approach (8 minutes).
- QCA (2 minutes).

#### Dataset Name

Insurance Company Benchmark (COIL 2000) – Insurance Customer/Claims Intelligence Dataset

#### Primary Link

[COIL 2000 – UCI Machine Learning Repository](#)

#### Alternative Links

[COIL 2000 Dataset – Kaggle](#)

COIL 2000 Dataset – OpenML

#### Format

TXT / CSV

#### Key Fields

The dataset contains customer/insurance attributes such as:

- Customer\_Subtype
- Number\_of\_houses
- Avg\_size\_household
- Customer\_main\_type
- Income
- Contribution\_car\_policies
- Contribution\_fire\_policies
- Contribution\_boat\_policies
- Contribution\_social\_security\_policies
- Contribution\_life\_insurance
- Contribution\_private\_accident\_insurance
- Number\_of\_car\_policies
- Number\_of\_fire\_policies
- Number\_of\_boat\_policies
- Number\_of\_life\_insurance
- Number\_of\_private\_accident\_insurance

#### Reason

The dataset contains insurance-customer and policy information that can be used to demonstrate **insurance customer segmentation, policy analysis, similarity-based retrieval, anomaly/risk analysis and AI-generated insurance insights**. It is a useful dataset for building the analytical foundation of an insurance intelligence assistant.

